# SYNCLAB: Centralized Workshop & Laboratory Allocation Platform

## Sequence and Activity Diagrams for Critical System Flows

**System Specification Document** | **Implementation:** C++17 / Qt Widgets / SQLite | **Scope:** Core Behavioral Modeling

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## Flow 1: Student Allocation Submission & Conflict Resolution (UML Sequence Diagram)

Figure 1 documents the synchronous interaction path when a student submits a reservation request. The system executes client-side business rule validation, initiates atomic slot conflict checking through the abstract database repository, and confirms persistent reservation integrity.

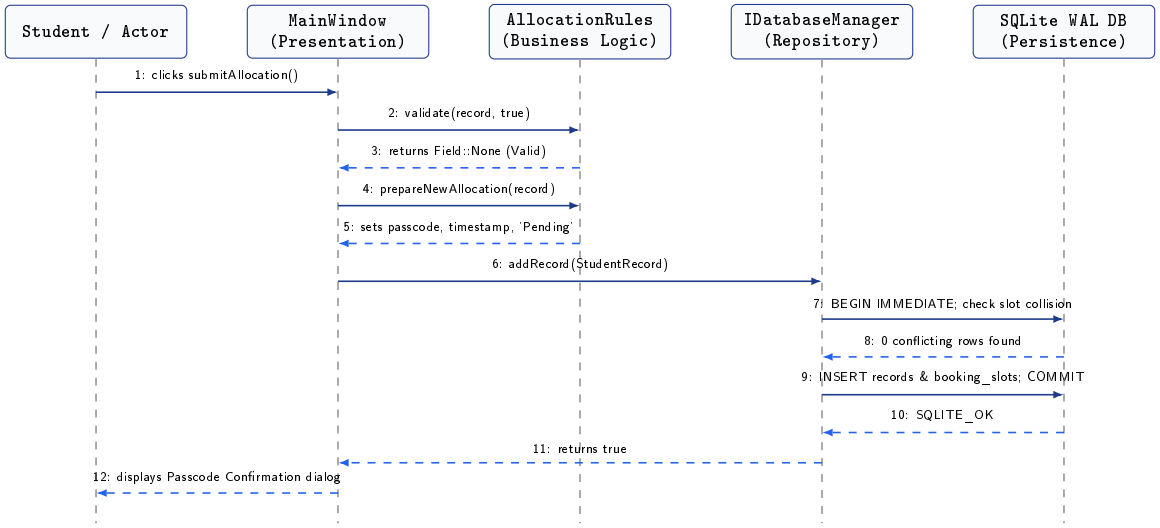



Figure 1: UML Sequence Diagram: Student Booking Validation and Atomic Persistence

## Flow 2: Administrative Review & State Transition Lifecycle (UML Activity Diagram)

Figure 2 models the complete operational activity lifecycle for administrators reviewing submitted records, handling filtering, applying status decisions, updating atomic remarks, and triggering notification dispatch.

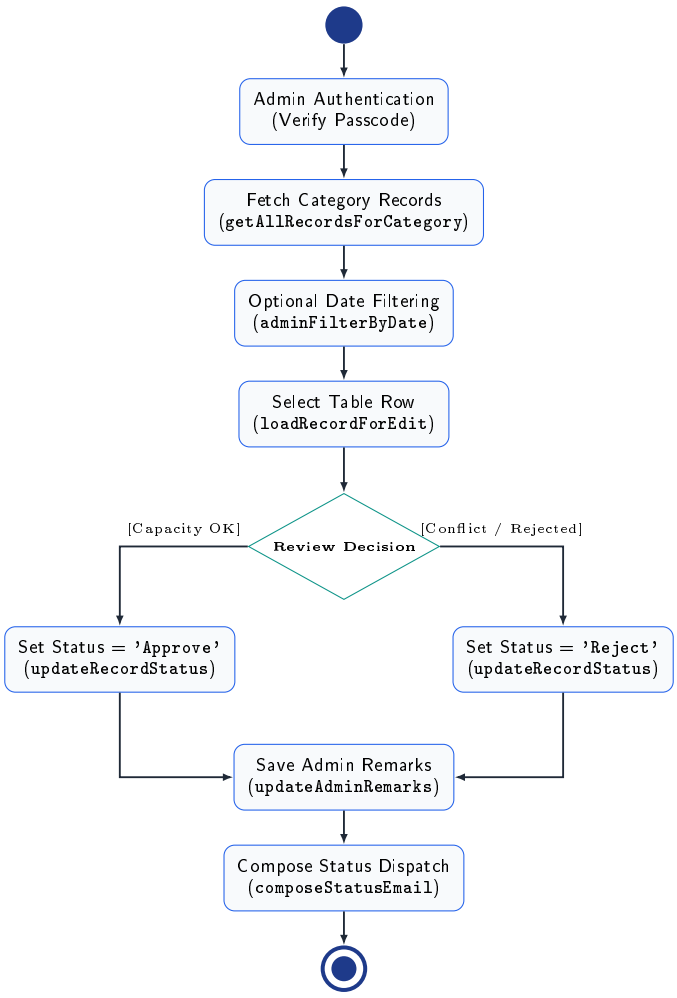



Figure 2: UML Activity Diagram: Administrative Review, Status Transition, and Notification